



Complex Social &
Computational Systems

THE DATA SCIENCE PIPELINE

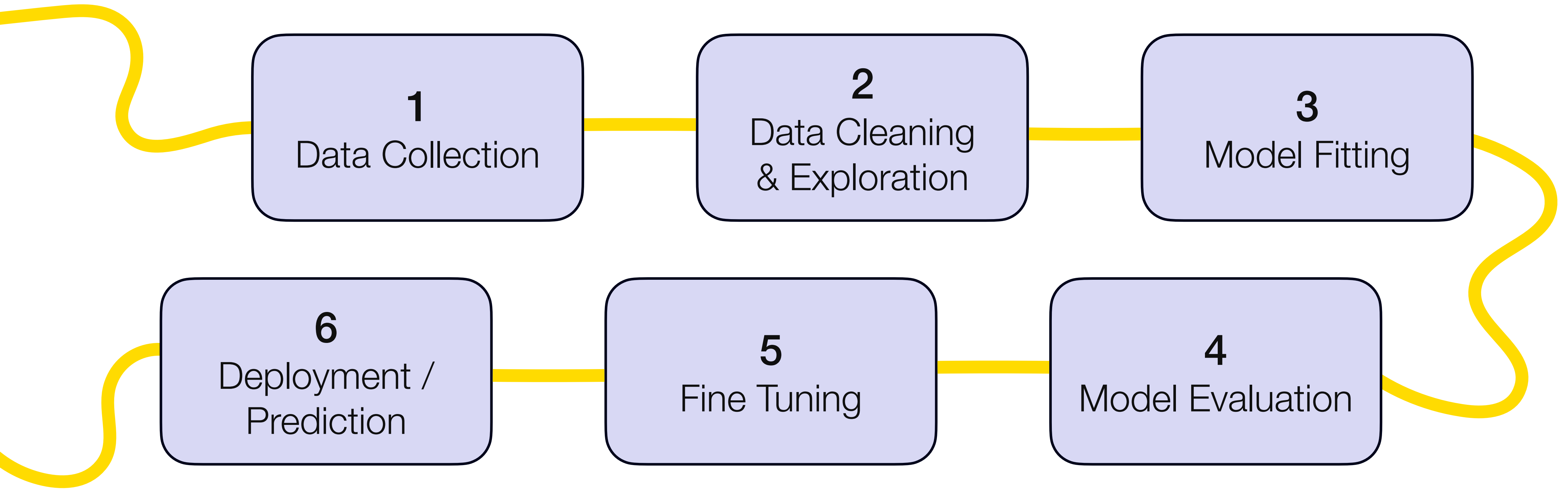
A BLUEPRINT FOR MACHINE LEARNING

A PSYCHOLOGIST'S GUIDE TO MACHINE LEARNING

ZURICH, MARCH 16-20, 2026

**A DATA SCIENCE PIPELINE ENCOMPASSES
THE PROTOTYPICAL STEPS REQUIRED TO
SUCCESSFULLY IMPLEMENT A MACHINE
LEARNING MODEL.**

A DATA SCIENCE PIPELINE USUALLY COMPRISES SIX STEPS.



1. DATA COLLECTION

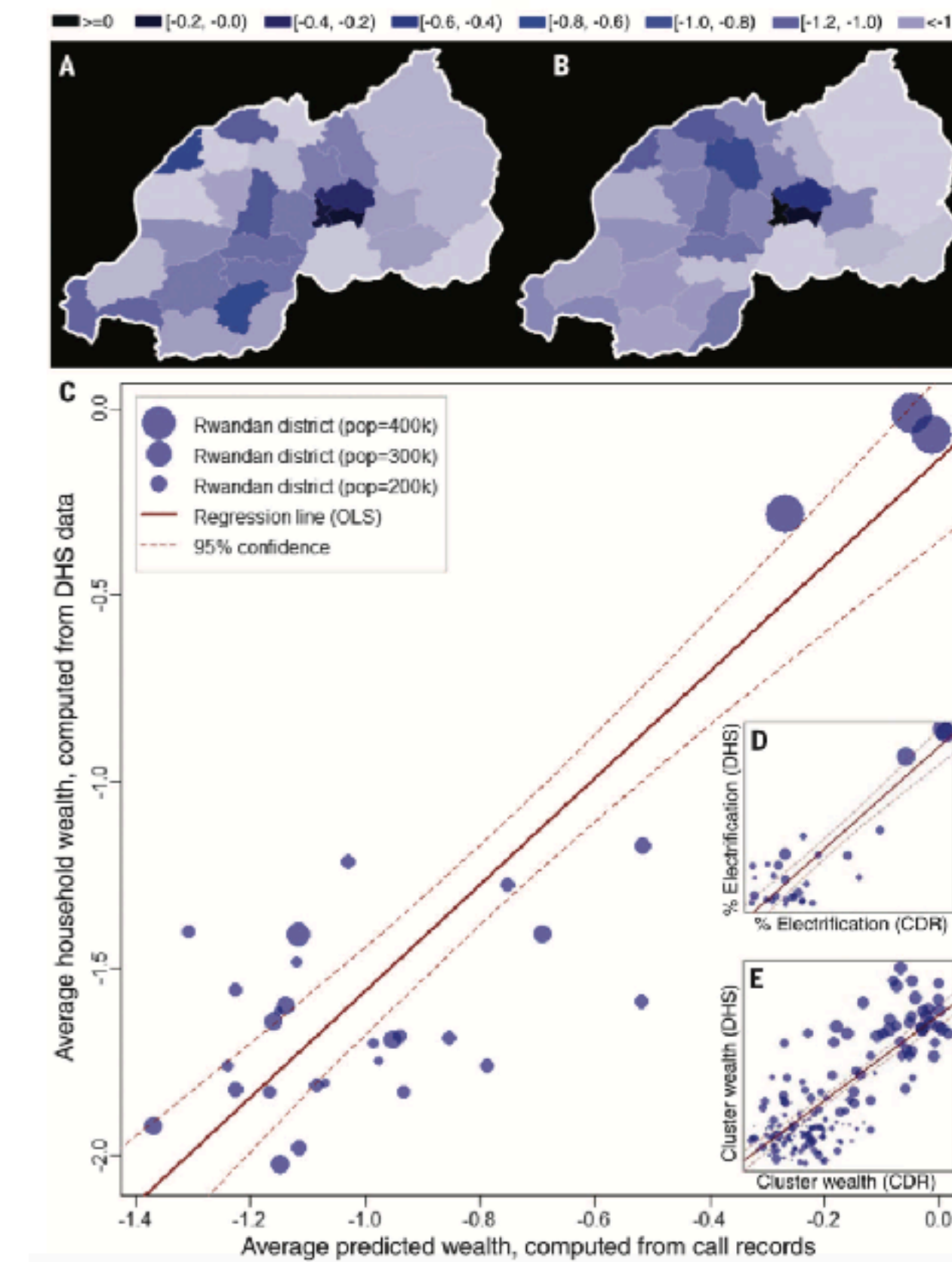
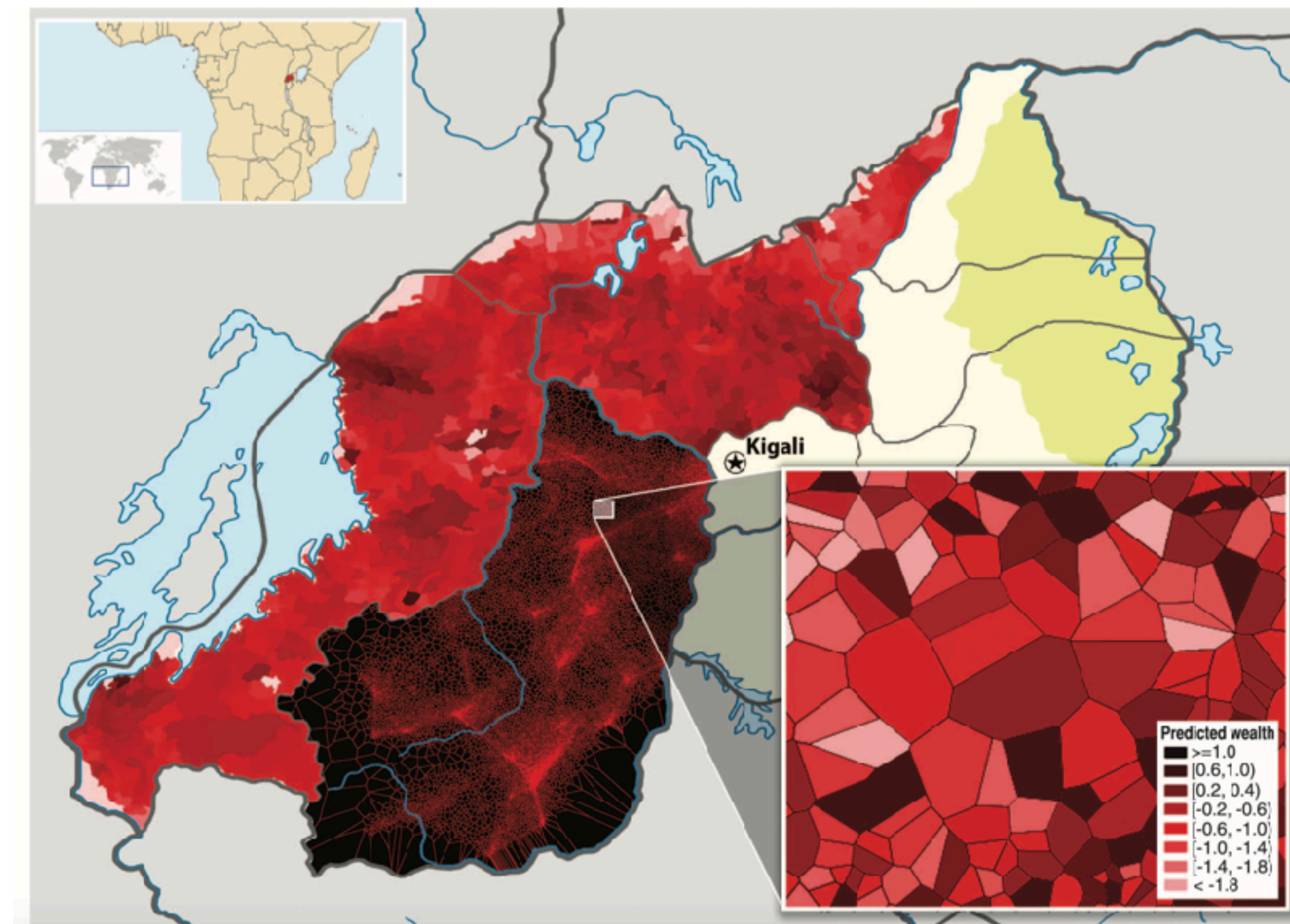
1 // DATA COLLECTION

**CUSTOM
MADE**



**READY
MADE**

1 // EXCURSUS: COMBINING CUSTOM MADE AND READY MADE DATA.



1 // EXCURSUS: CUSTOM MADE DATA ON A LARGE SCALE.

PROJECTS

ABOUT

GET INVOLVED

TALK

BUILD

SIGN IN

REGISTER

FROM THE ORGANIZATION: GALAXY ZOO

ABOUT

CLASSIFY

TALK

CO

Galaxy Zoo

ENGLISH

Galaxy Zoo recently migrated onto Zooniverse's new architecture, as announced on our blog. For details, see [Zooniverse's blog post here.](#)

GALAXY ZOO

Few have witnessed what you're about to see

LEARN MORE

FROM THE ORGANIZATION: GALAXY ZOO

GET STARTED!

You can do real research by clicking to get started here!

6% COMPLETE JWST COSMOS

MESSAGE FROM THE RESEARCHER

For more than 15 years Galaxy Zoo volunteers have helped astronomers make sense of the beautiful complexity of galaxies in our Universe. We appreciate each and every one of your classifications.

GALAXY ZOO

By the numbers

139,600

Volunteers

74,995

Subjects

1,492,337

Classifications

4,709

Completed subjects

<https://www.zooniverse.org/projects/zookeeper/galaxy-zoo>

1 // EXCURSUS: MAKING READY MADE DATA USABLE.



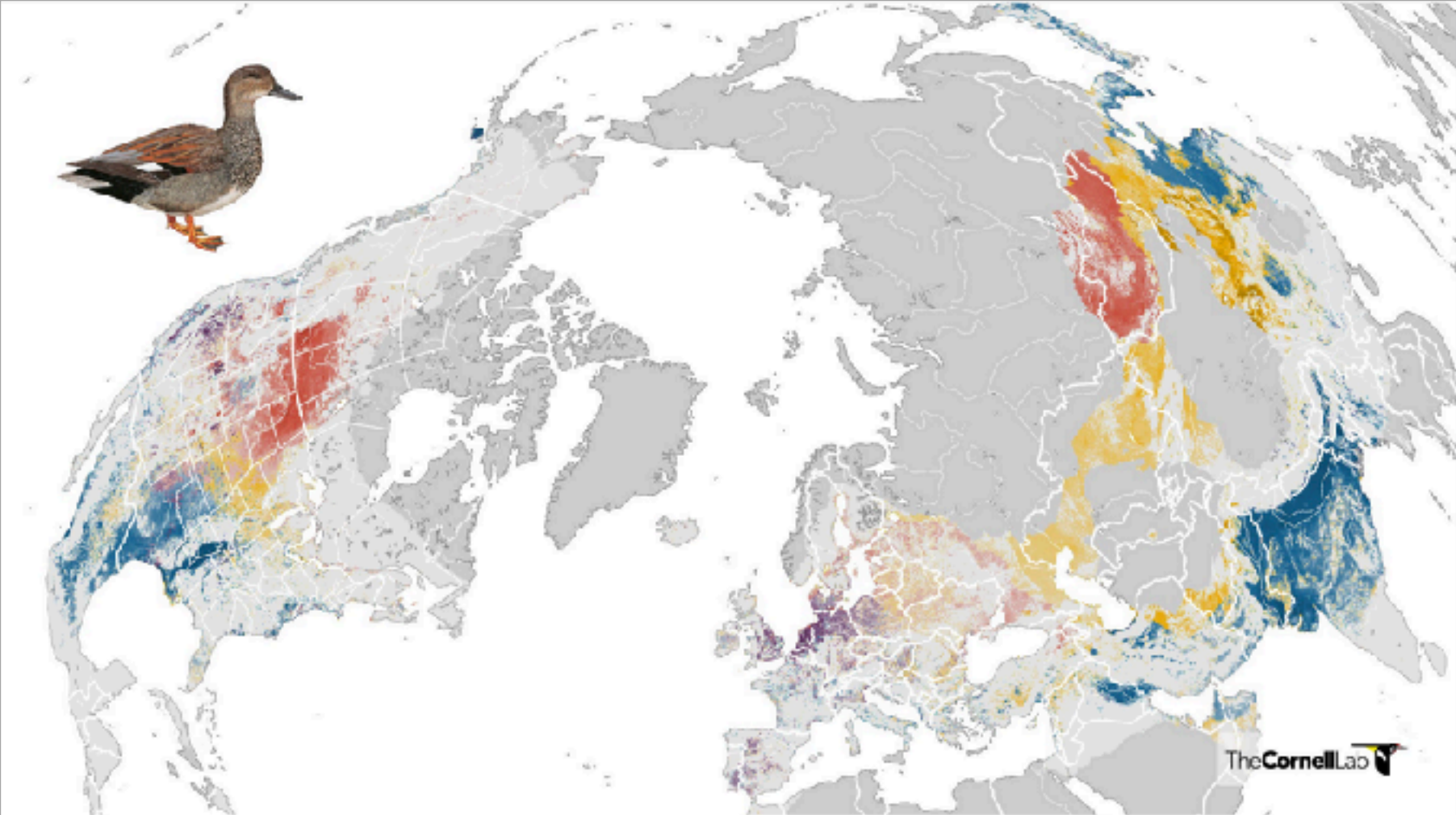
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Mallard *Anas platyrhynchos* © Emma Navaz



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1 // EXCURSUS: HOW MUCH DATA DO I ACTUALLY NEED?

IT DEPENDS.

On the model. On the problem to solve. On the quality of data.

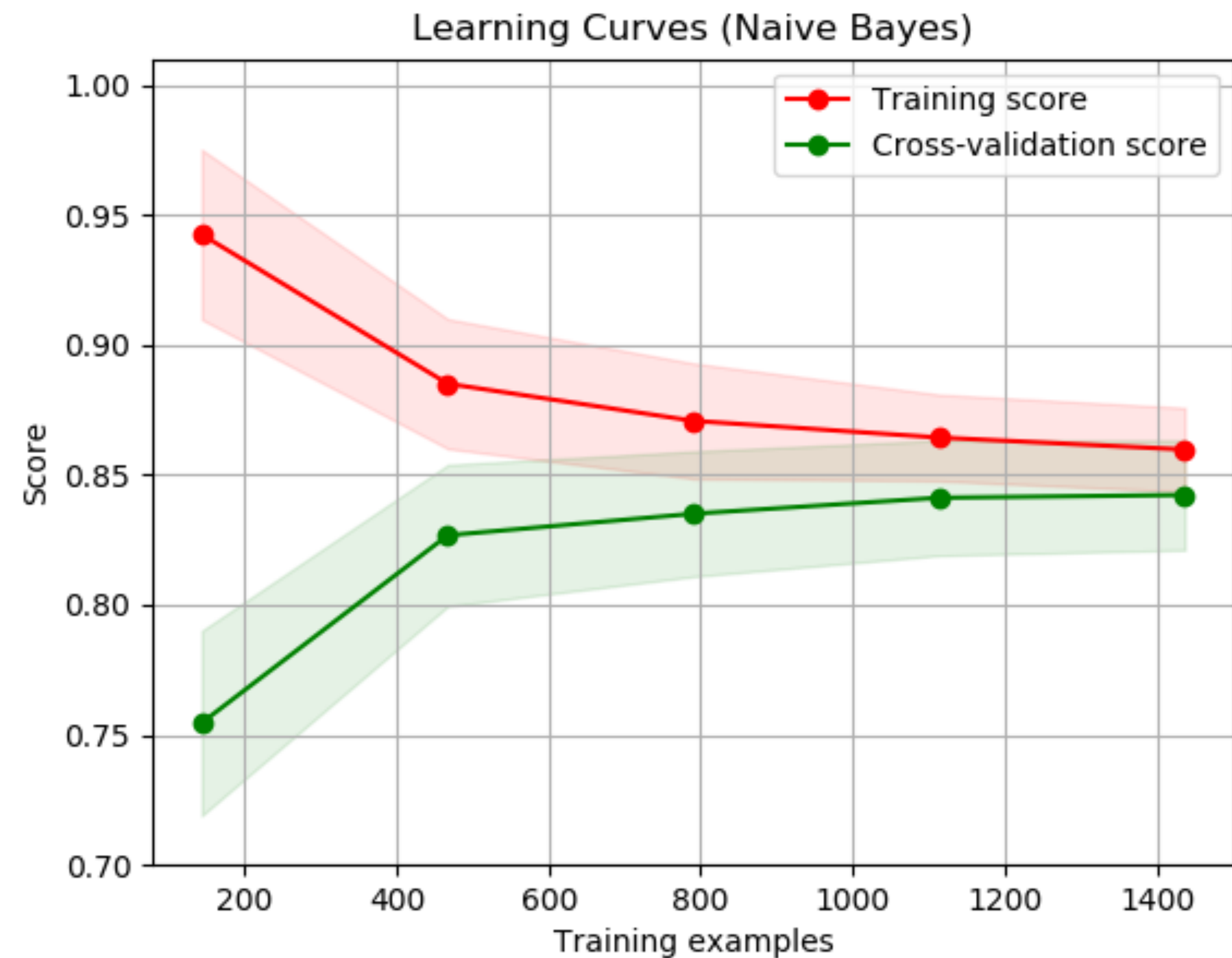
More parameters, more data.

More classes, more data.

Clean data, less data. Messy data, more data.

Ambiguous concepts, more data.

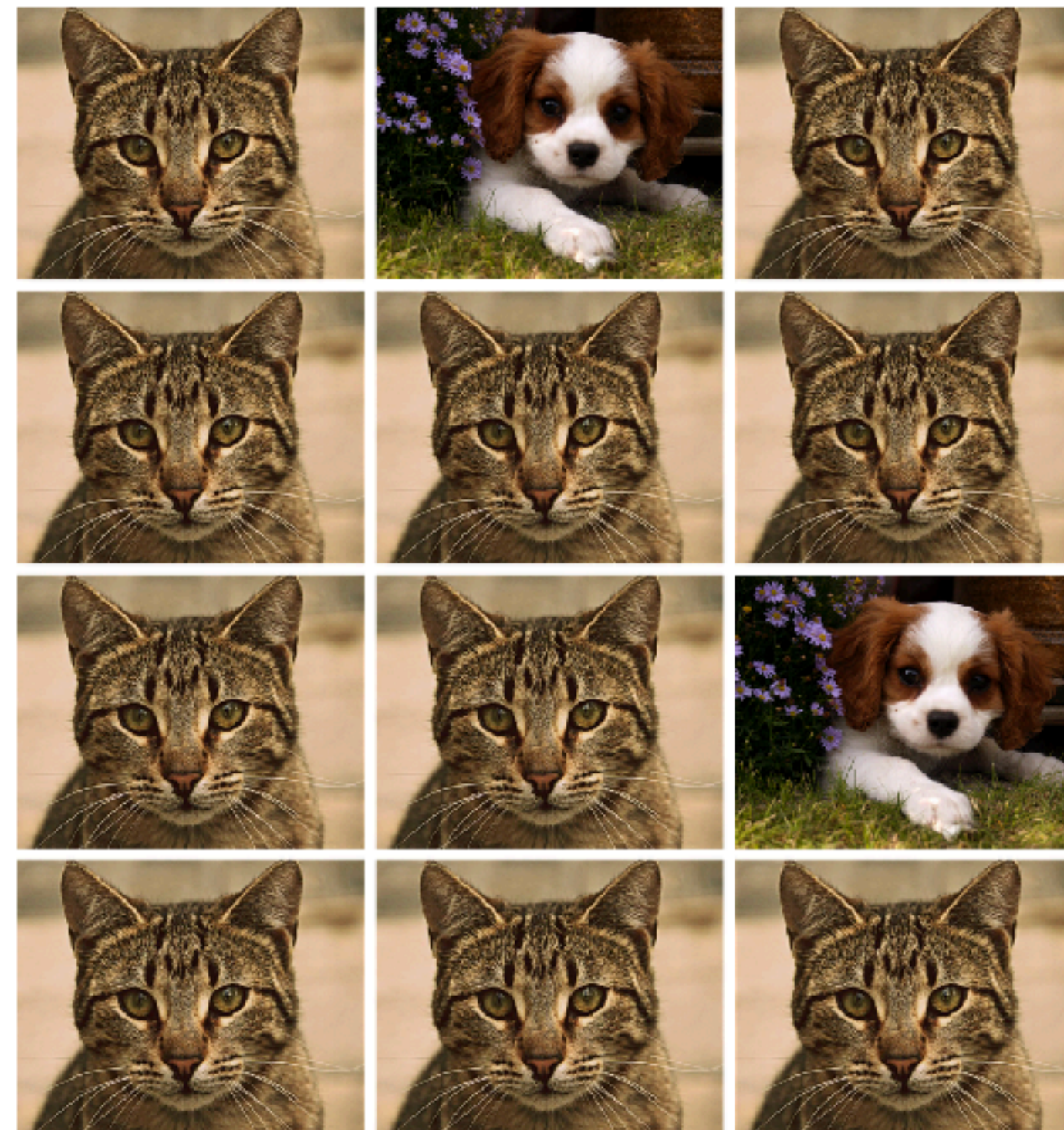
1 // EXCURSUS: HOW MUCH DATA DO I ACTUALLY NEED?



Orient yourself on similar problems in the literature and analyze the **learning curve**.

1 // EXCURSUS: WHAT IS HIGH QUALITY DATA?

Data without
BIAS in the
training sample.



Training Data



Unknown Data



Data with
VARIANCE in the
training sample.

2. DATA CLEANING

2 // DATA CLEANING

Randomize

Deduplicate

Normalize

Impute

Data Augmentation

Dimensionality Reduction

Feature Engineering

2 // EXPLORATORY DATA ANALYSIS

Data Visualization

Descriptive Statistics

Plausibility Checks

Pattern Recognition

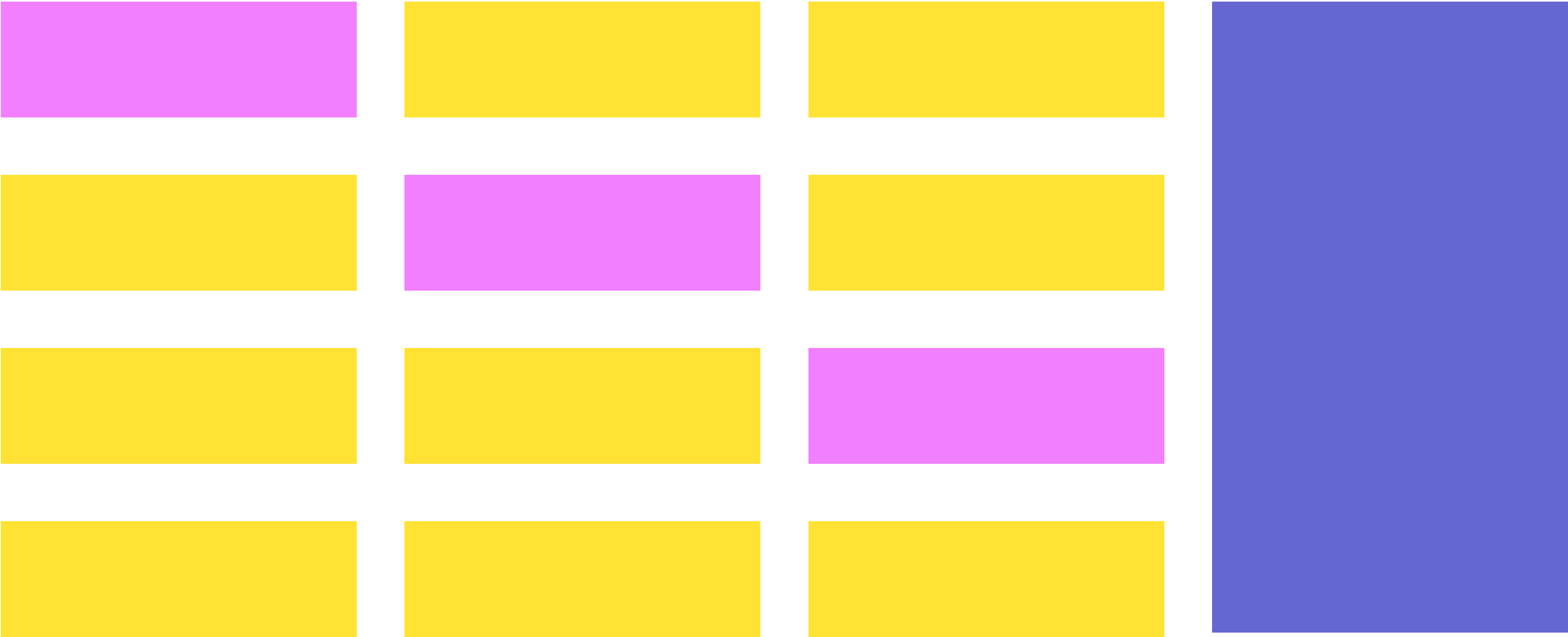
BEFORE MODEL TRAINING: PREPARING FOR CROSS-VALIDATION

TRAINING SET: is used to estimate the model parameters

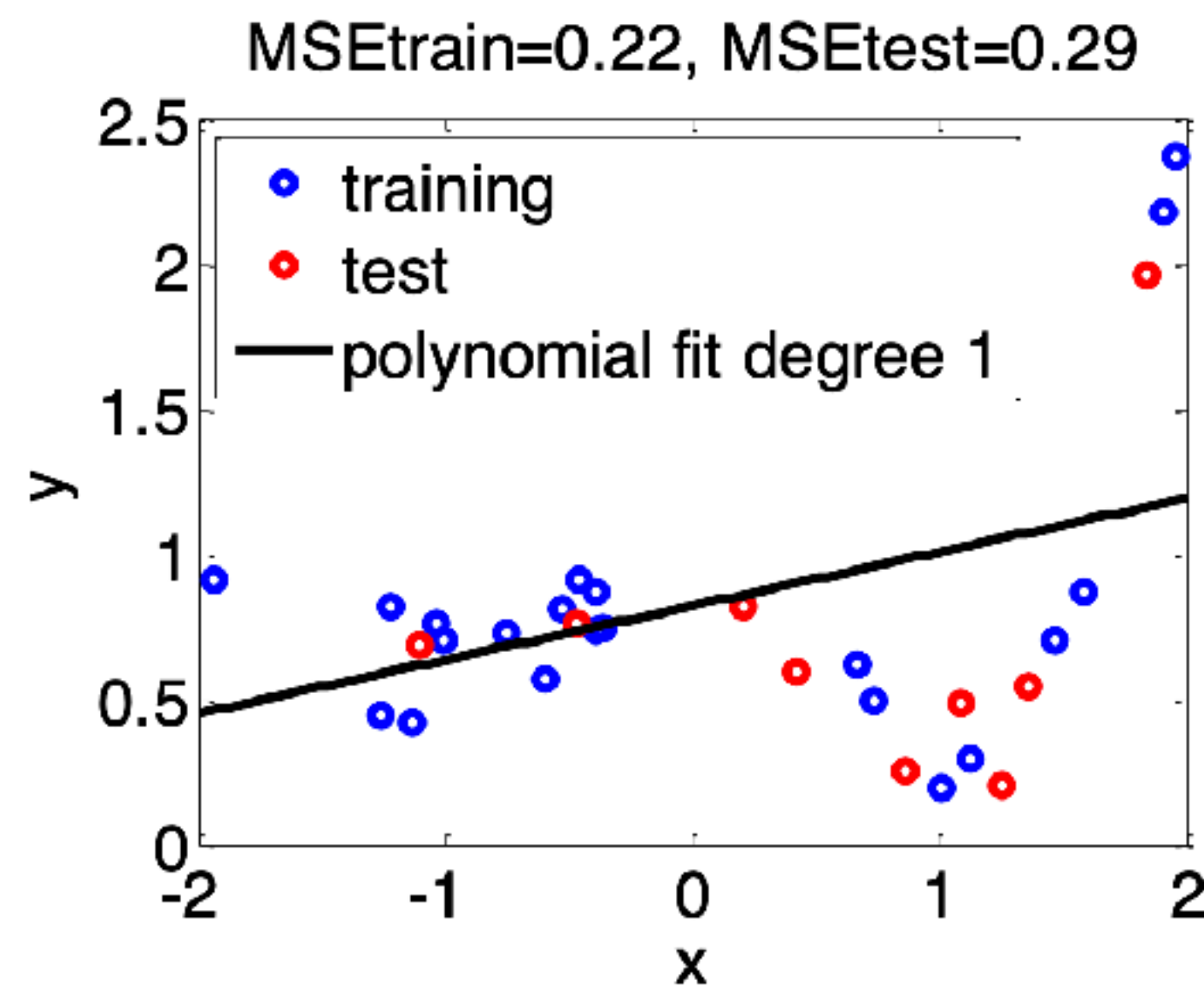
VALIDATION SET: is used to test the quality of the model during the training phase; the model with the best evaluation results on the validation set is selected.

TEST SET: is used to test the actual performance of the model on unseen data

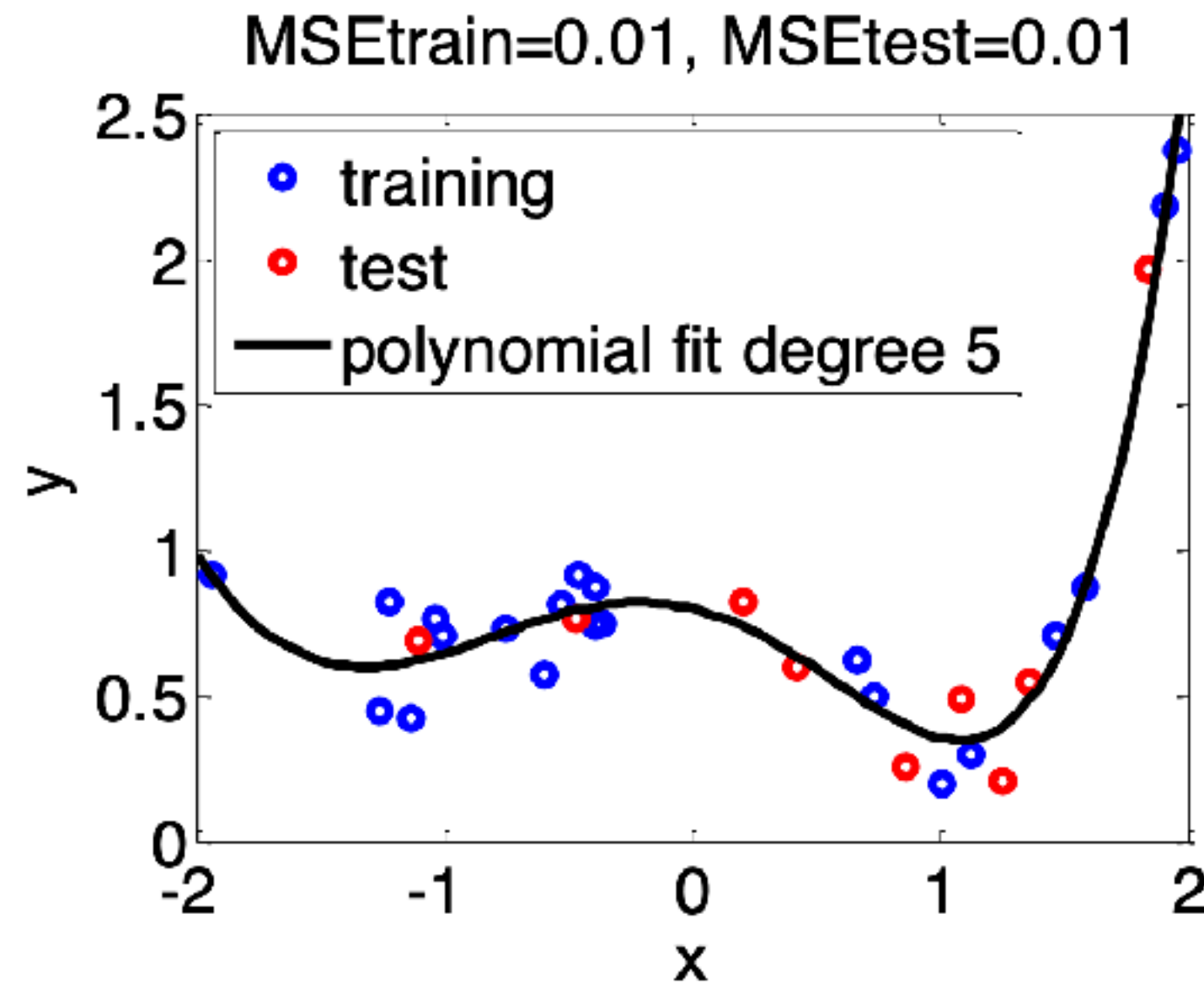
BEFORE MODEL TRAINING: PREPARING FOR CROSS-VALIDATION



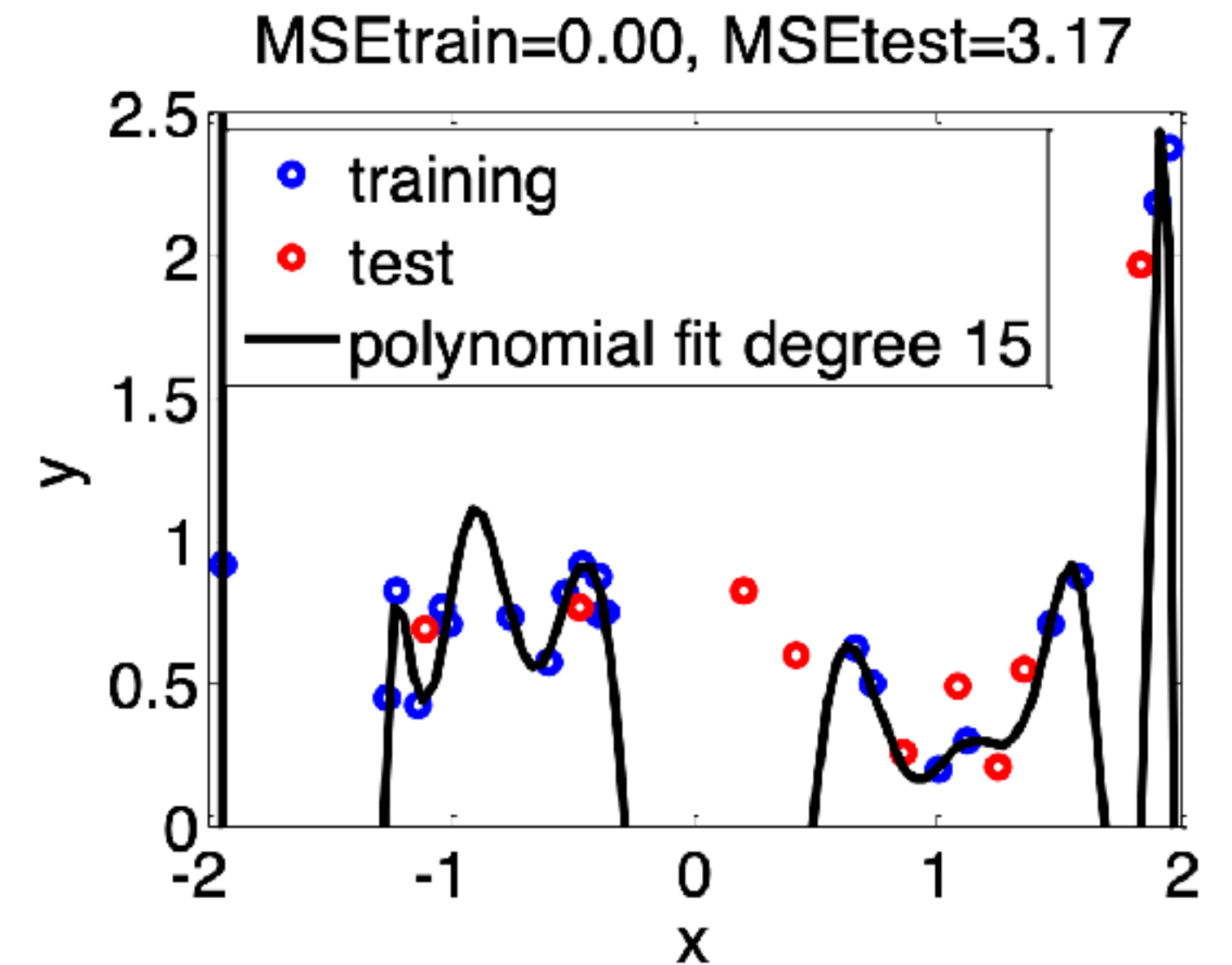
CROSS-VALIDATION PREVENTS OVER- AND UNDERFITTING.



underfitting

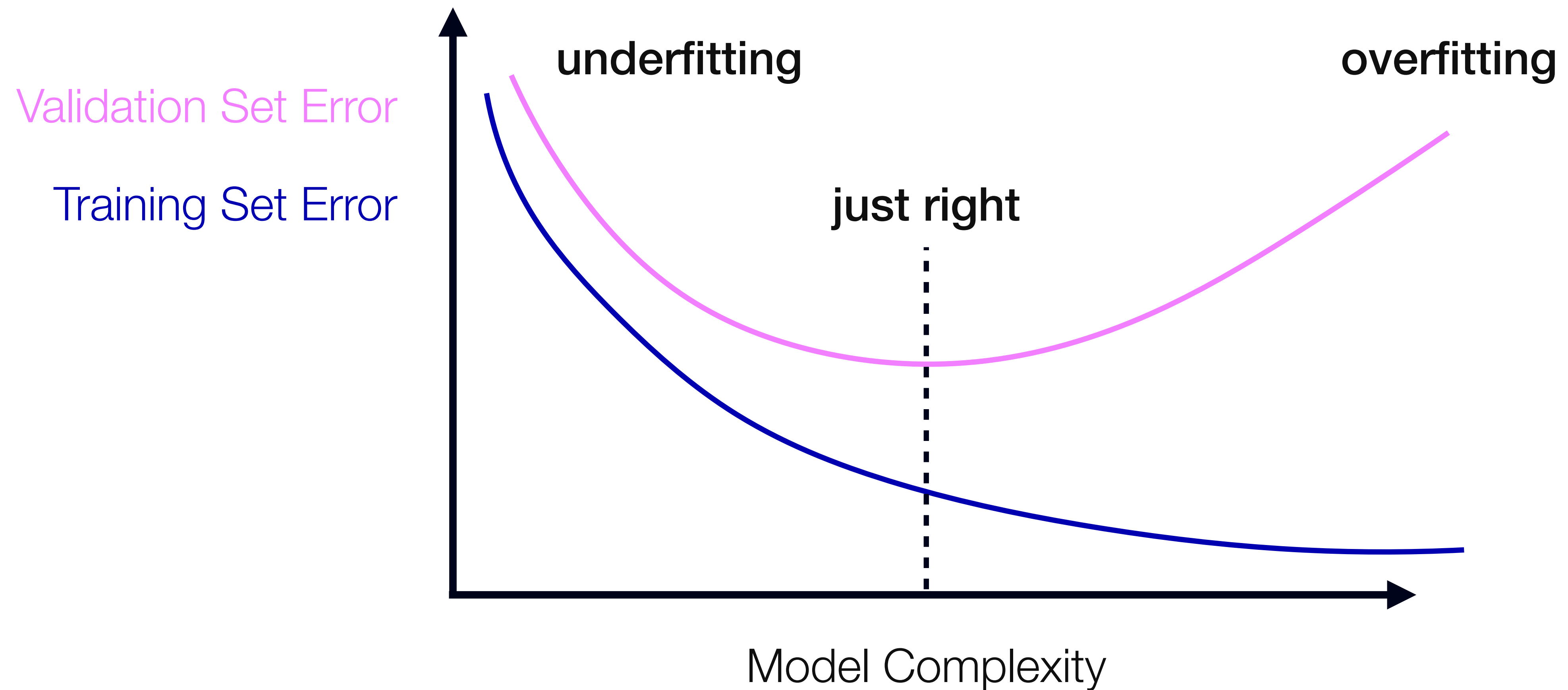


“just right”

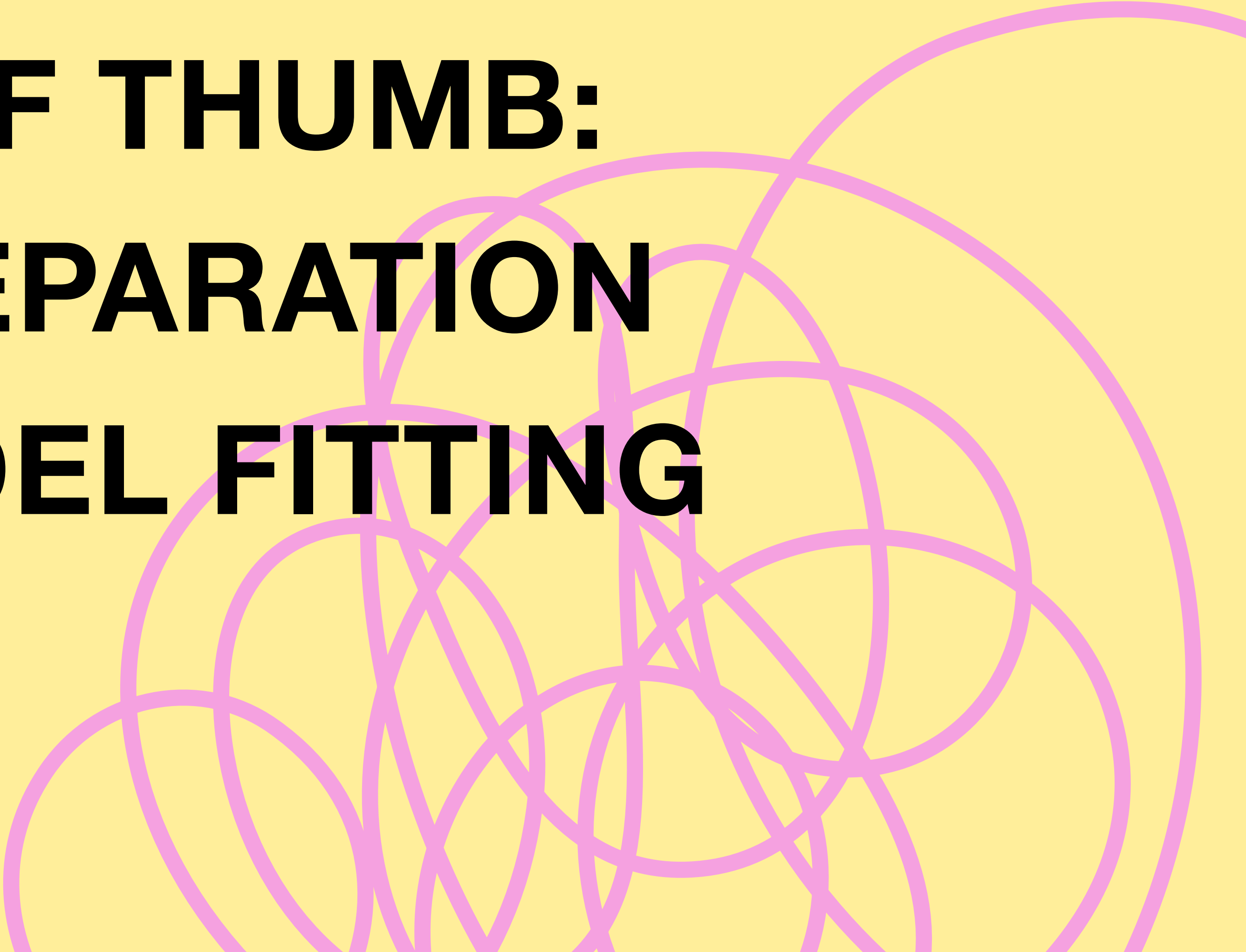


overfitting

CROSS-VALIDATION PREVENTS OVER- AND UNDERFITTING.

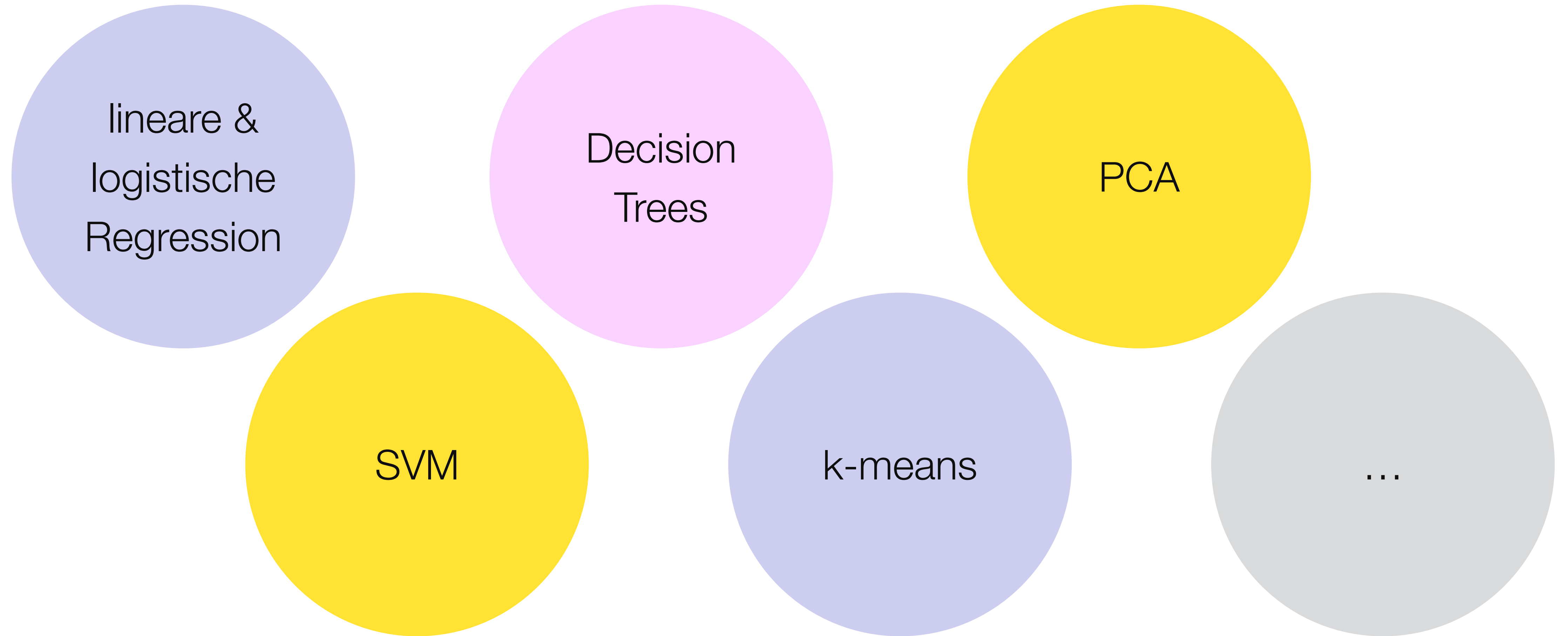


RULE OF THUMB:
80% PREPARATION
20% MODEL FITTING



3. MODEL FITTING

3 // MODEL FITTING



4. MODEL EVALUATION

4 // MODEL EVALUATION

Knowing an algorithm's performance is important at three points:

1. To **guide training** towards better performance.
2. To **stop training** when performance no longer improves.
3. After training ends, to assess the algorithm's **final performance**.

4 // MODEL EVALUATION FOR REGRESSION TASKS

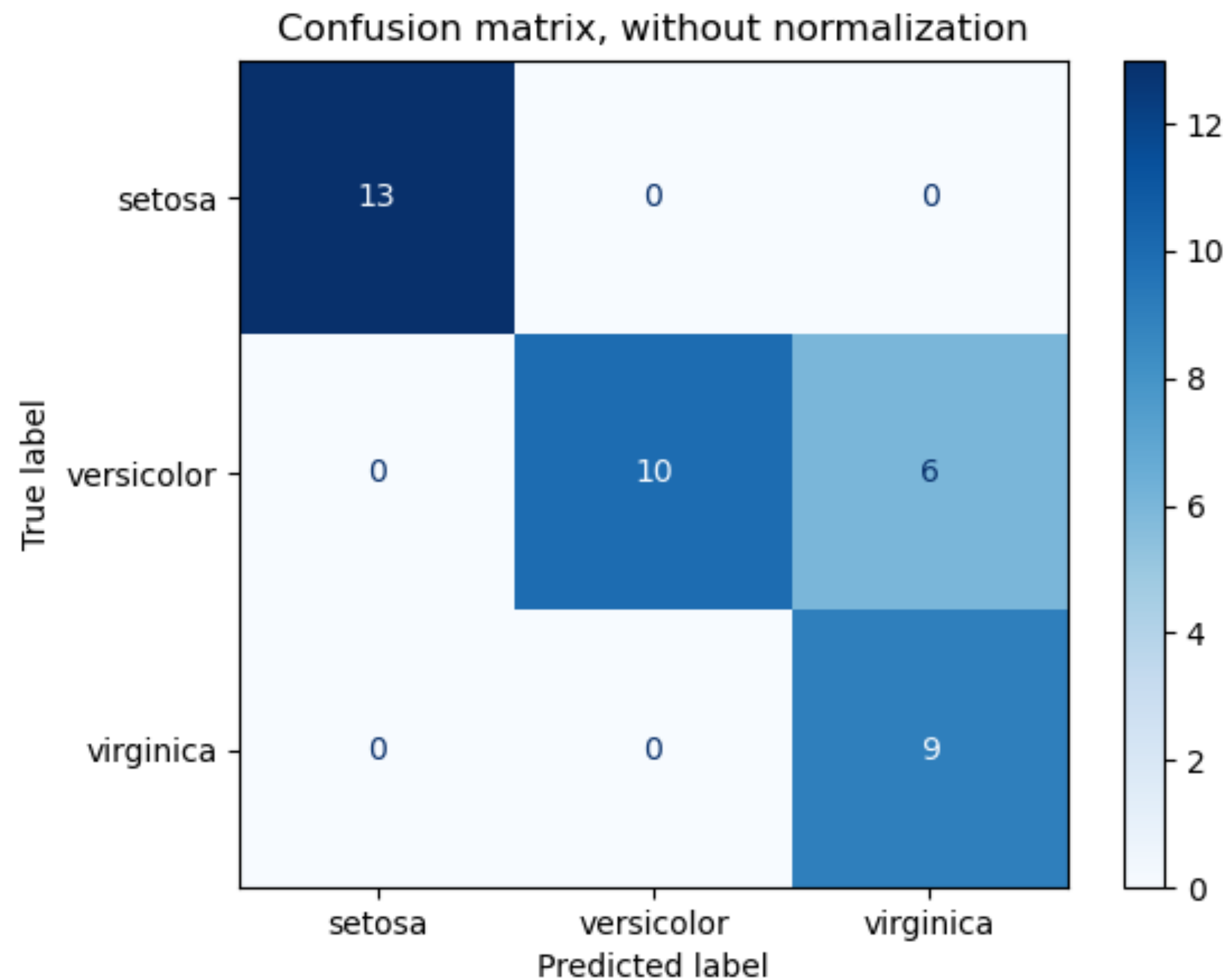
$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

Mean Squared Error

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}$$

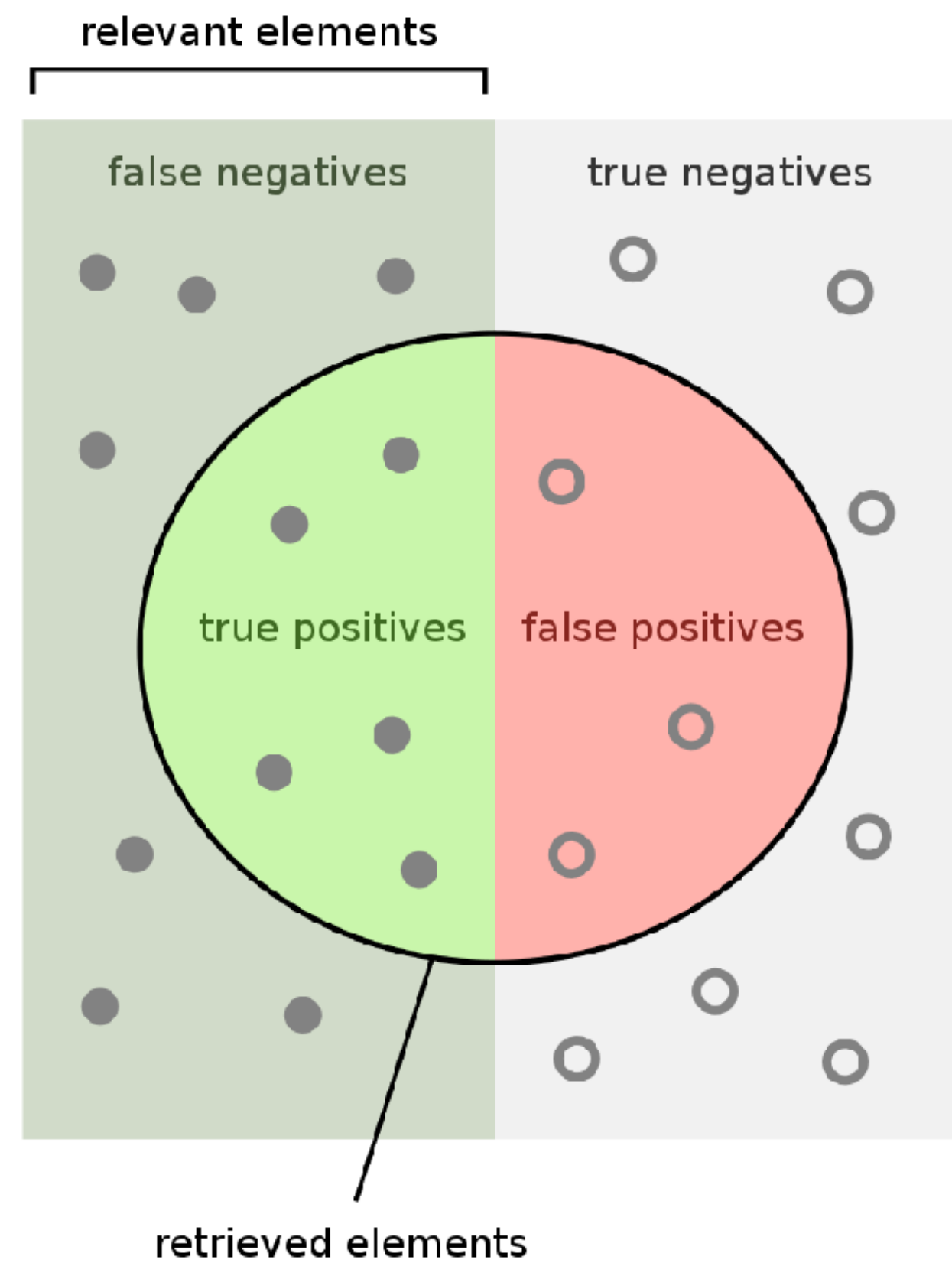
Coefficient of
Determination

4 // MODEL EVALUATION FOR CLASSIFICATION TASKS



**THE
CONFUSION
MATRIX**

4 // MODEL EVALUATION FOR CLASSIFICATION TASKS



How many retrieved items are relevant?

$$\text{Precision} = \frac{\text{true positives}}{\text{true positives} + \text{false positives}}$$

How many relevant items are retrieved?

$$\text{Recall} = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}}$$

$$F1 \text{ Score} = 2 \times \frac{\text{recall} \times \text{precision}}{\text{recall} + \text{precision}}$$

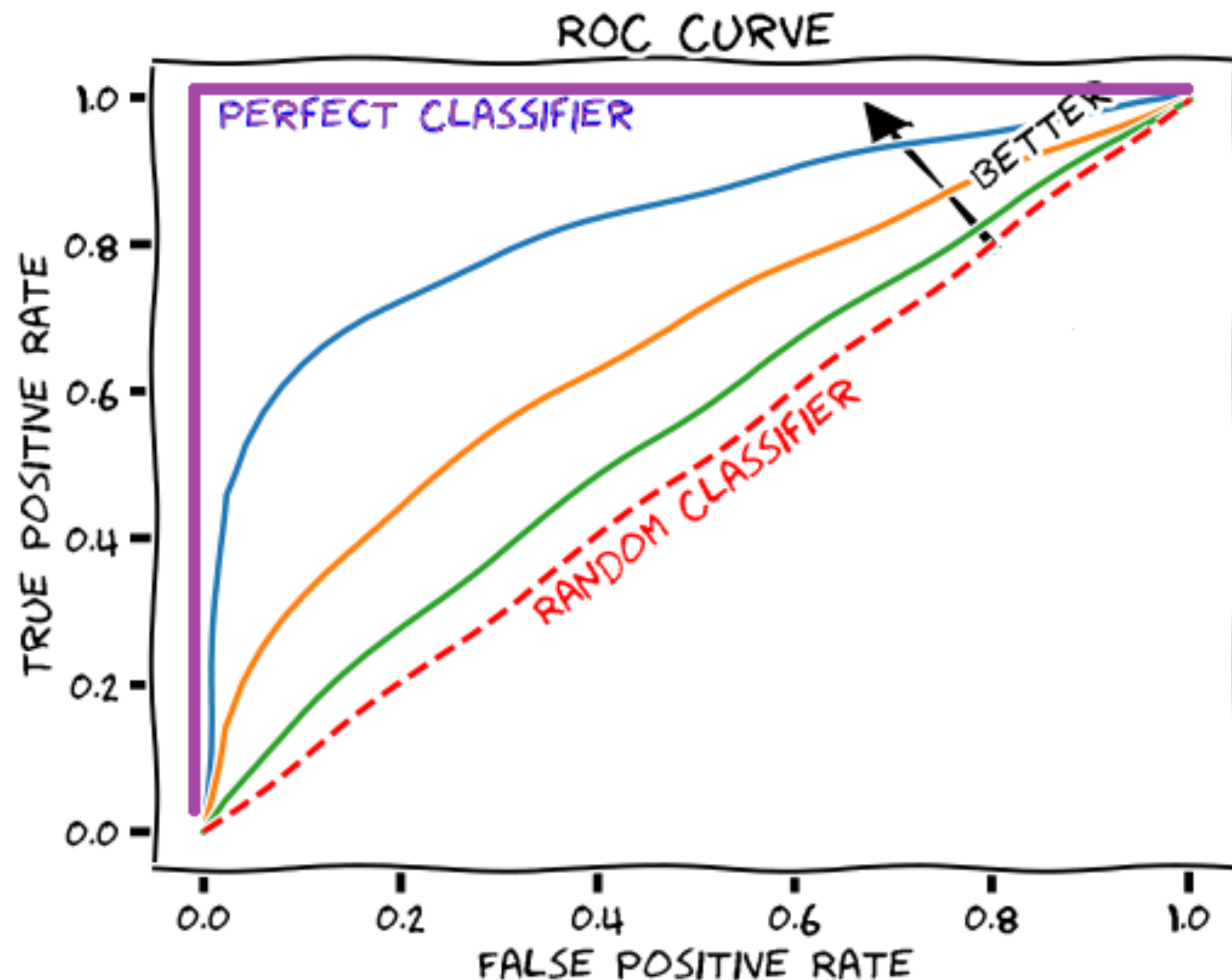
THE
F1 SCORE

4 // MODEL EVALUATION FOR CLASSIFICATION: WHEN TO USE WHAT?

Different indicators satisfy different application cases:

- **Recall** is best when classes are not equally frequent and false negative errors are particularly severe.
- **Precision** is best when classes are not equally frequent and false positive errors are particularly severe.
- **F1 score** is best when classes are not equally frequent and both false positive and false negative errors are severe.

4 // MODEL EVALUATION FOR CLASSIFICATION TASKS



ROC-AUC CURVE

Receiver Operating Characteristics
Area Under the Curve

5. FINE TUNING

5 // FINE TUNING

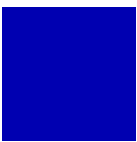
Adjusting the hyperparameters of the model.
(e.g. the number of training steps or the learning rate)

TRIAL AND ERROR!

5 // FINE TUNING: THE RANDOM GRID SEARCH

LR \ epochs	5	10	20
0.01			
0.005			
0.0025			
0.0001			

 selected

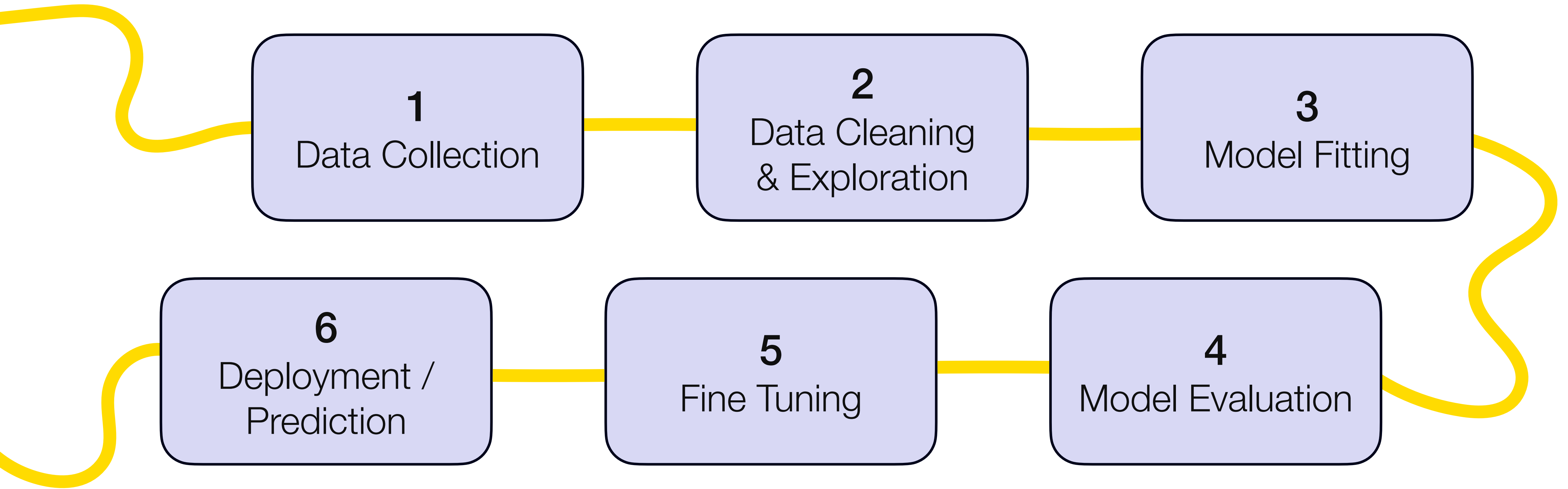
 not selected

EXCURSUS: BENCHMARKING



6. PREDICTION

A DATA SCIENCE PIPELINE USUALLY COMPRISES SIX STEPS.



16:15 - 17:00

**MODEL EVALUATION
EXERCISE**